Claim sheet (gate 1, draft for Navish, 2026-11-13)

Thesis. Models read and quote; code decides; what the file leaves open becomes a held request with one question.

Every claim below names its figure, the evidence that exists today, and the evidence that must be produced. “Exists” means stored and reproducible; the numbers come from the figure drafts’ data files. Study 3’s figures are for its 96 sealed tenancy claims unless theft is named.

Question 1. What happens to a fact the file leaves open?

#	Claim	Figure	Exists today	Must be produced
1	With only “true” and “false“, a reading must take one branch: it asks for that branch’s documents too early or drops the other’s. A third value lets code request what both branches need and hold the rest.	3	Formal: follows from the definition of the workflow; shown on a real fragment of a public rule set.	Nothing.
2	One sentence in the file decides one line of the checklist, and the line can be followed back to the quoted words.	1, 2	One stored pair of claims, drawn as it is. **Not true in general yet:** in 144 stored theft pair-runs the core changed only the governed line in 12; in 130 another line moved too, because readings of other facts differ between runs.	A fix that makes the same text give the same checklist. Then sealed twins in Study 4: two files one sentence apart, exact and collateral changes counted for every design.
3	Of 100 requests that depend on an open fact, CasePath holds almost all, each with the right question. A prompt that is not told to hold asks for half of them too early. A prompt that is told holds them too, but then also holds or drops requests the file settles.	4	On the cost-efficient model the core holds 99.6 of 100, the plain prompt 48, the prompt that may hold 97 (86 documents in 32 claims). Of 100 settled requests the core gets 96 right, the prompt that may hold 90 (481 documents). **Lost today:** on the frontier model the core holds 87, the prompt that may hold 98. The prompt that may hold ran after the test.	Study 4, with needless holds counted as errors.

Question 2. Does CasePath beat the strongest prompt as cases get real?

#	Claim	Figure	Exists today	Must be produced
4	CasePath’s error holds its level as files get longer, rule catalogues larger and open facts more frequent; the strongest prompt’s error rises.	5, main table	**Nothing that supports this.** Study 3’s files have 77 to 320 words. The public-rule benchmark has run no baseline. Today a prompt told the hold rules ties CasePath on tenancy (0.35 against 0.31 strict errors per claim) and beats it on claims that deny their defining event (0.00 against 0.61 on the frontier model).	Three fixes on development data, then Study 4 on fresh sealed files. Every cell is shown, won or lost.
5	The lead does not cost many times a prompt, and the same file gives the same checklist when run again.	7	The core costs about 4 times the prompt that may hold on the same model (USD 0.0030 against 0.0007 per claim) for 0.73 against 0.89 errors; on the frontier model it costs 3.4 times as much and errs more (0.70 against 0.58). As tested it would cost about 40 times that prompt on a 2,500-word file (extrapolated). The same checklist in all three stored runs: 70 of 96 claims for the core.	Fixes to cost (the file read once and reused; supplied documents judged in one call; one reading where it settles the fact). Ten runs of the same file in Study 4.

Question 3. What does an organisation gain when its rules change, or when it adds a kind of claim?

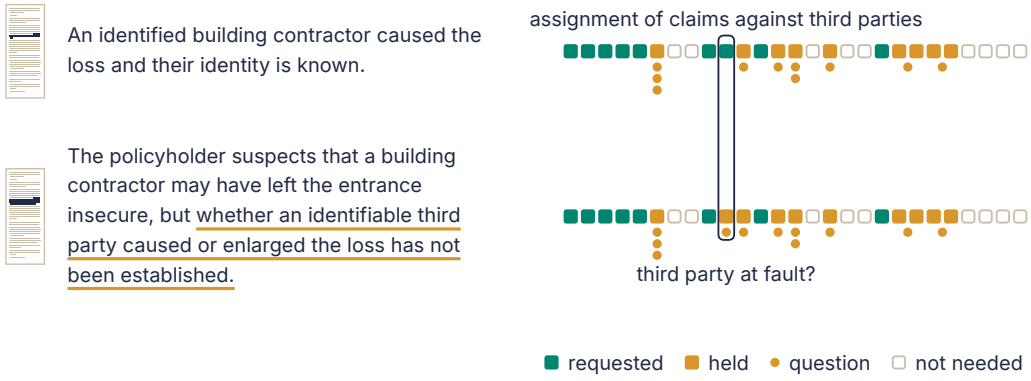
#	Claim	Figure	Exists today	Must be produced
6	After a rule edit, code recomputes every stored checklist with no model call and changes only the lines the edit reaches. A prompt must read every file again.	6	Replay of stored readings under one edited rule: 15 of 421 lines changed in 56 theft files, 14 as the edit requires, no model call. The prompt’s side is not measured.	Rule-change experiment: a set of edits; exact changes, collateral changes and model calls for every design. An edit that adds a new fact costs CasePath readings of that fact only; this is measured too.
7	A new kind of claim is added from public text in six steps by a session that has not seen the system.	table	Three regimes were built this way (177 cited rules); effort was not recorded; first use found 39 wording errors, since repaired.	A fresh session adds a fourth regime; effort is recorded; its sealed result enters Study 4.
8	The system is small: a workflow is a data file and a new rule is one line; routing; several attachments and scans; presumptions; Knowledge from handler corrections, replayed without a model.	one listing or frame each	In the product at the tested commit.	Frames on synthetic cases, requested from the product session.

What the paper does not claim

No claim rests on the hackathon dossiers or the expert study. Studies 1 and 2, the commit timeline and the deviations list go to the supplement. If a cell of the main table is lost after the fixes, the table shows it and one sentence says why.

Figure 1. One changed sentence

The reader should see that replacing the one sentence that states a fact with one that leaves it open changes one checklist line from “requested now” to “held, with one question“, and leaves every other need in the same place and state, because that line is the only need whose rule names the fact the sentence leaves open.



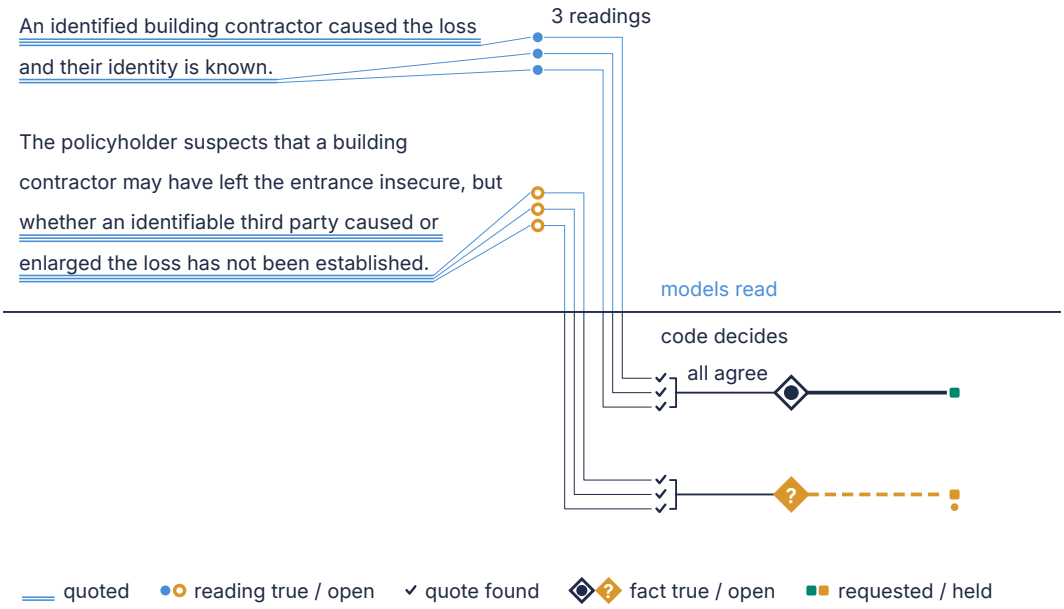
One stored run: two claims, one sentence apart; the second leaves the third party’s role open. Of 27 needs, one turns from requested to held. Each dot: one question per open fact a held line awaits. Nothing else changed in 12 of 144 stored pair-runs.

Status: Measured: the two sentences and their page marks, the 27 marks per claim and their states, the question dots (attachment formal, from required_when), the amber underline on the words quoted as open. Formal: the frame on d15. Not yet measured: no element.

Rejected construction: b, one page and one fused strip: it shows a computed diff, not the two stored records; it draws one file for two and drops the questions.

Figure 2. The journey of one sentence

The reader should see that the model contributes only quoted words and a verdict per reading, and that the quotation check, the agreement of the three readings, the fact's value deciding its rule, and the checklist line are computed by code, because the checklist line can be followed back across the one rule, along three lines, to the words each reading quoted.



One sentence, stated (top) or left open (bottom). Three readings quote the file. Below the line, code checks each quotation, admits a value only when all three agree, and the fact decides its rule: request now, or hold with a question (marks as Figure 1).

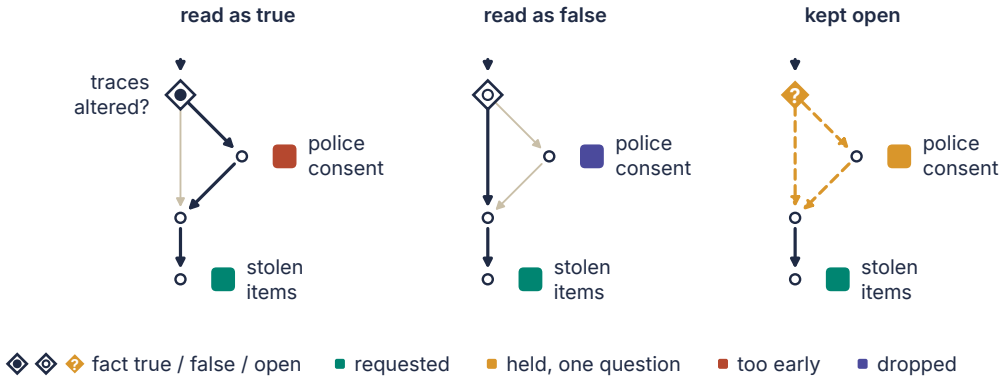
Status: Measured: sentences and words, quotation underlines, reading marks, ticks (recomputed by substring), the fact's value, the checklist line. Formal: agreement bracket (team.py rule, measured inputs), diamond node, outgoing edge. Schematic: horizontal positions. Not yet measured: no element.

Rejected construction: a, one lane, the open sentence alone: no partner; amber can't be read as a change without the stated lane.

Figure 3. Why “open” needs its own value

The reader should see that a reading that may only say true or false must send the case down one branch, and so either asks for the police consent too early (true) or drops it (false), while the value “open” lets code request the list of stolen items, which both branches need after they rejoin, and hold only the police consent, with one question, because the two branches of “traces altered?” meet again before the list is needed.

«Ob er das inzwischen gemacht hat oder ob alles noch so ist wie am Parkplatz, weiss ich nicht.»
Whether he has had it repaired yet, or all is still as it was at the car park, I do not know.



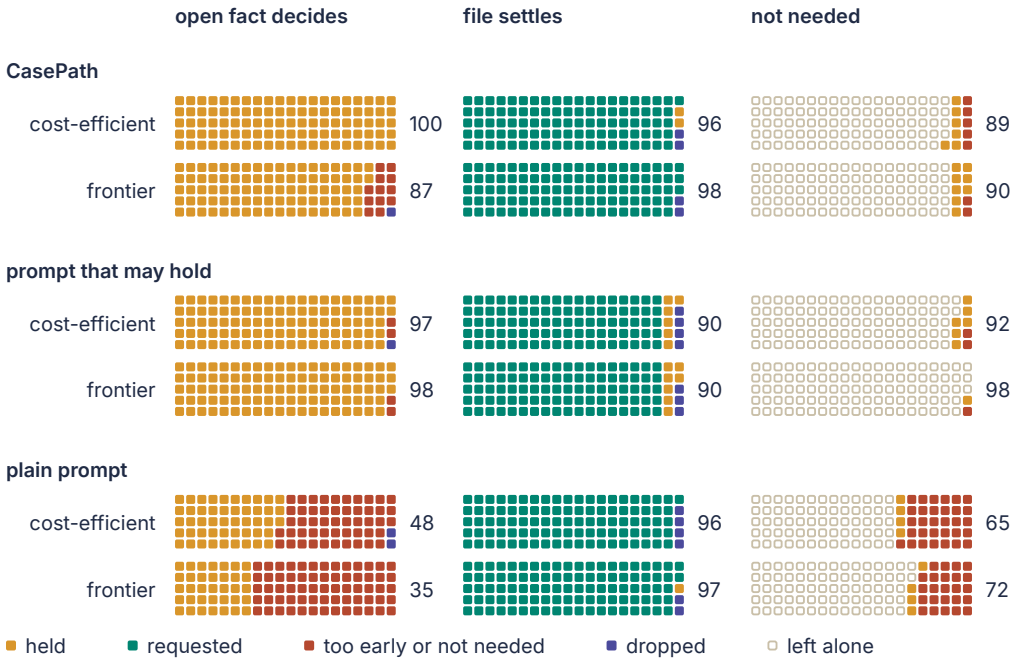
One file, read three ways (diamond filled: true; hollow: false; amber: open). Read as true, the police consent is asked too early; as false, it is dropped. Kept open, code requests the stolen-items list, needed after the branches rejoin, and holds only the consent.

Status: Formal: follows from the definitions, computed by the product’s own evaluator, on a real instance. Measured: all three stored readings answer open, quoting the sentence. Schematic simplification: a_instruction contracted. Not yet measured: no element.

Rejected construction: b, the fragment once beside a 2 × 3 matrix: the mechanism (branch walked, rejoin) is not visible per reading; it hides the cause.

Figure 4. The fate of 100 open facts

The reader should see that a plain prompt asks too early for half of the requests an open fact decides, that a prompt told the hold rules holds them as CasePath does but then strays where the file settles the request, and that CasePath's own stray marks sit where nothing is needed, because each design is judged on all three states of the reference.



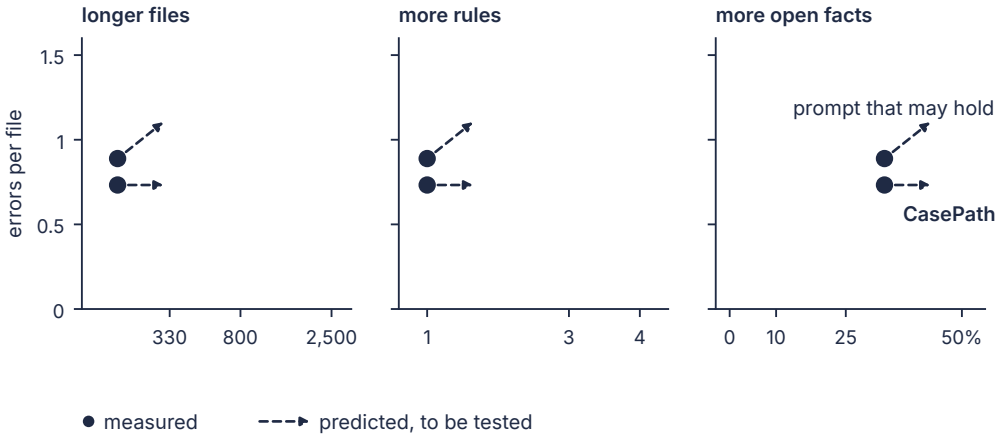
Each block is 100 checklist lines of one reference kind; colour is what the design did. The plain prompt asks too early; the prompt that may hold strays where the file settles; CasePath strays where nothing is needed. Circle: cost-efficient model; square: frontier.

Status: Measured: every mark. Formal: grouping into 5 × 20 blocks and largest-remainder rounding. The frontier CasePath and prompt rows rest on one run. Not yet measured: no element.

Rejected construction: b, continuous bars: loses the line as unit, erases slivers.

Figure 5. As cases get real

The reader should see that today each design is measured at one point only (short files, one regime's rules, a third of the files open), and which direction the registered test predicts for each as files get longer, rules more numerous and open facts more frequent, because a design that reads one fact at a time and decides in code should hold its level where a single prompt does not.



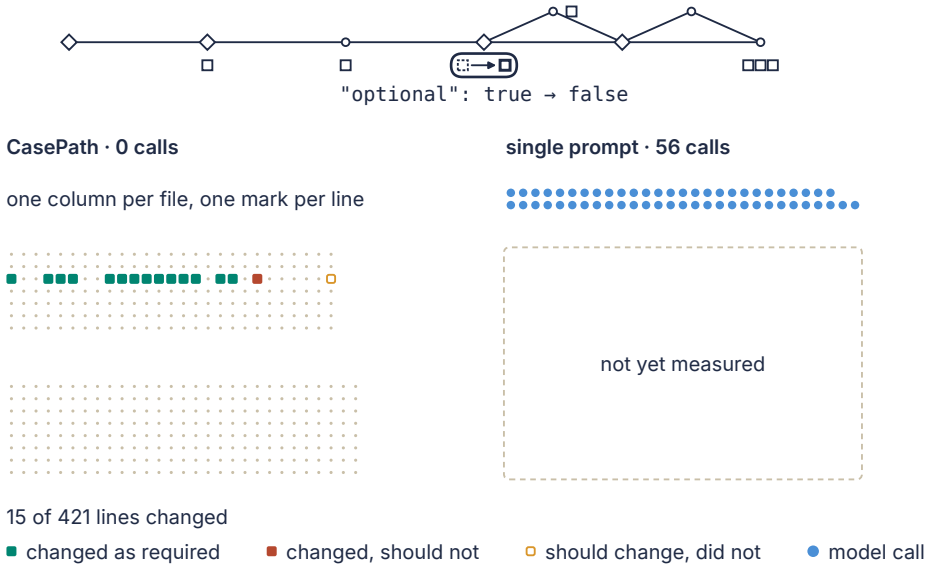
Each design has one measured point today (dot): Study 3's short files (median 172 words), one regime's rules, a third of files with an open fact. Arrows show what the registered test predicts, not data: CasePath level, the prompt rising. Cost-efficient model.

Status: two measured points per panel (Study 3); the arrows are the registered prediction, not data. This figure has no content until Study 4 runs.

Rejected construction: b, dots alone: the hypothesis then lives only in the caption and the panels look like a finished result at one point.

Figure 6. A rule changes

The reader should see that after one edited rule line, code recomputes every stored checklist of the regime from the stored readings with no model call and changes only the lines the edit reaches, while a prompt must read every file again, because CasePath stores one reading per fact and the rules are data.



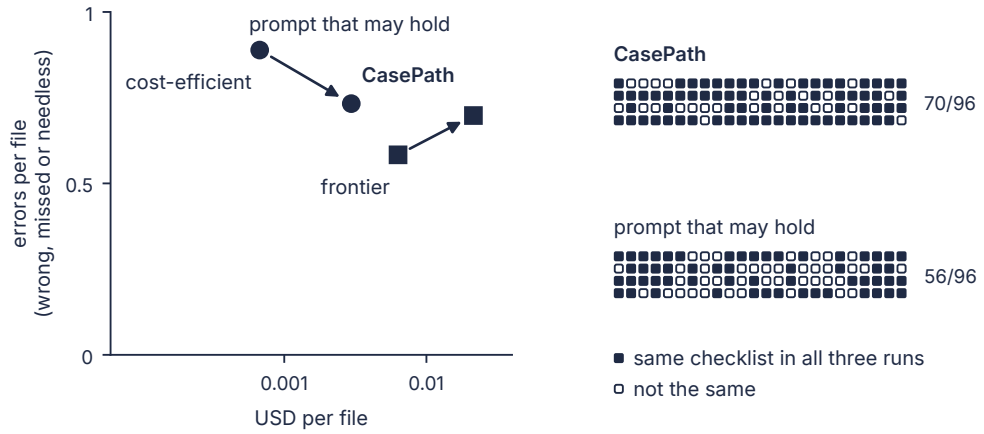
One line made the police report required (ringed). From stored readings, code recomputed every checklist: blocks are theft away (21 files) and theft at home (25); 10 files were never routed. Rust: a report on file whose finding was rejected; amber outline: an unrouted file.

Status: Measured: stored replay, the changed lines, 0 model calls. Formal: the prompt reads every file again, 56 calls. Not yet measured: the prompt's changed lines (dashed frame).

Rejected construction: b, a before→after transition table: loses which file and which document changed, and the graph.

Figure 7. The price

The reader should see that on the cost-efficient model CasePath costs about four times the prompt that may hold and makes fewer errors, that on the frontier model it costs more and makes more, and that the same file gives the same checklist in all three runs more often with CasePath, because a lower error is only useful at a price and when it repeats.



Arrows lead from the prompt that may hold to CasePath: on the cost-efficient model 4.4 times the cost buys 0.16 fewer errors per file; on the frontier model 3.4 times buys 0.12 more. Run three times, CasePath repeats 70 of 96 checklists, the prompt 56.

Status: Measured: all marks (cost from ledgers; errors and repeats from three stored runs). Not yet measured: ten runs (Study 4).

Rejected construction: b, three stored checklists of the median file as strips: identical for both designs, true for the median, but it hides the 26 and 40 claims that change.

Costed plan for the fixes and Study 4 (gate 1, for Navish, 2026-11-13)

Nothing below has been spent. Costs are model credits in USD from a cost model built on the stored ledgers (`rebuild/plan/COST_MODEL.md`, `STUDY4_COST.md`; the design is data in `study4_cost.py`). Every figure for files longer than today's (575 words, four attachments) is an extrapolation; the low figure is the central estimate, the high one adds 25% and lets output grow with length. Work by writing and checking sessions costs no credits.

1. What the figure drafts found, and what it changes

1. **The hold can be taught to a prompt.** On Study 3's sealed tenancy claims the prompt that may hold holds 97 of 100 requests that depend on an open fact; CasePath's core holds 99.6 on the cost-efficient model and 87 on the frontier model. The difference today is on the other side: of 100 requests the file settles, that prompt holds or drops 10, the core 4, and the core's needless holds sit on claims that deny their event. The paper's first question is answered by all columns of Figure 4, not by one.
2. **"Nothing else moves" is not true yet.** One changed sentence moved exactly one line in 12 of 144 stored theft pair-runs of the core; in 130 another line moved too, because readings of other facts differ between runs. The same theft text gave the same checklist in all three runs for 8 of 128 texts (tenancy: 70 of 96).
3. **Three readings are not better than one.** In the core, one reading gives 0.55 deviations per tenancy claim, three give 0.74: the extra readings add needless holds.
4. **Cost is not in the re-reading.** Reading again only where the first reading does not settle a fact would save 6 to 7% per file. 62% of the core's cost is the judging of supplied documents, and each of about ten calls carries the whole file. On a 2,500-word file the core as tested would cost about 40 times the prompt that may hold.
5. **After a rule edit, the stored readings are enough.** One edited line changed 15 of 421 stored checklist lines with no model call, 14 of them exactly as the edited rule requires. Read again without any edit, CasePath's checklist lines differ as often as a prompt's (about 6 against 7%): the gain is from not reading again.

2. Fixes, on development data only

#	Fix	Aimed at	Measured on development files by
a	A quoted denial of the defining event counts as a denial	needless holds on denied events (finding 1)	errors on the 24 denied-event files
b	One reading; a second and third only where the first is not a quoted true or false	findings 3 and 4	errors and cost per file against three readings
c	The file is the fixed first part of every call, so later calls read it from the cache; supplied documents judged in one call	finding 4	cost per file by length; target at most three times the prompt that may hold on a long file
d	The same text gives the same checklist: fixed sampling where the provider allows it, and a stored reading is reused when the quoted words are still in the file	finding 2	identical checklists over five runs; lines moved in one-sentence twins
e	Quotation matching that tolerates scan noise; the dismissal routing question	first use of the benchmark	holds caused by unmatched quotations; routing failures
f	Rule conditions from cited text with a "not this" line	the two over-broad conditions	done: the lint flags 0 of 47 conditions of the public rule sets

Every baseline gets what it can use: the corrected rules, the hold option, routing as a first step if it helps, the same budget for self-consistency. The strongest prompt per model is named from development data and frozen.

3. The benchmark, extended (no credits)

- **Length:** three levels: about 330 words with zero to four attachments (today's), 800 words with three attachments, 2,500 words with six attachments, a fifth of them scanned.
- **Rules:** the same files are read with the rules of one regime, of three regimes and of four regimes in one catalogue.
- **Open facts:** share of files with an open fact 0, 10, 25 and 50%.
- **Twins:** for some files a second file one sentence apart.
- **A fourth regime**, added from public text by a session that has not seen the system, following the six steps; its hours and sessions are recorded.
- About 930 sealed files are written by a model family that is not evaluated and checked blind, as before.

4. Study 4 (one registered test, after a freeze with an external timestamp)

- **Cells:** three regimes × two models × five designs (CasePath's core; the prompt that may hold; self-consistency at CasePath's cost; the model executing the rules; ReAct). The main table shows every cell.
- **Files per regime:** 160 for the open-fact sweep (40 per level, short); 40 medium and 40 long at the 25% level; the 40 short files of that level read again with the larger catalogues; 30 twins; 20 files run ten times on the cost-efficient model and three times on the frontier model. Fourth regime: 120 short files.
- **Measures:** all deviations per file against the clean-room reference (a needless hold is an error); the share of files with the same checklist in every run; cost per file.
- **Rule change:** three edits per regime. For every design: lines changed as the edit requires, lines changed that it does not reach, model calls. One edit adds a fact, so CasePath's cost for that case is measured too.
- **Hypotheses, per model, frozen in order before the run:** at the hard end of each axis (long files, four regimes, half the files open) CasePath makes fewer errors than the strongest prompt; at the easy end it makes no more (margin 0.15 per file); after the rule edits it changes more of the right lines, fewer of the wrong ones, with fewer calls. A lost cell is reported as lost.

5. Cost

	Cost-efficient model	Frontier model	Total
Development cycle (fixes, priced pilot of long files, three frontier gate rounds)			44 to 61
A. Sealed test as in section 4	56 to 72 with development	272 to 353	327 to 424
B. As A; on the frontier model half the files in every cell	56 to 72	148 to 193	203 to 264
C. As B; self-consistency and ReAct on the cost-efficient model only	56 to 72	82 to 110	138 to 182
A, if fix (c) halves the core's cost	33 to 42	169 to 220	202 to 262

Totals include the development cycle. The largest items are the core and its cost-matched self-consistency on long files on the frontier model (38 to 59 each).

6. Order, and what the paper can claim at each stop

1. **Development cycle, capped at USD 60.** Ends with a report: each fix's measured effect, the measured cost of long files, and which cells are won, tied or lost on development data against the strongest prompt. Claim after this stop: none new; the paper stays a mechanism paper with Study 3 and the fair-prompt check.
2. **Benchmark extension and the fourth regime** (no credits), in parallel with 1.
3. **Freeze** (artefact, baselines, plan, order of hypotheses, generator seed), timestamped outside our control, at least three days before the run. You choose A, B or C here, with the pilot's prices in hand.
4. **Study 4, once.** Claim after this stop: what comes out, cell by cell.
5. **Cold-reader test of the figures, then the 18 pages.**

7. Decisions I need

1. Approve the development cycle (cap USD 60) and the benchmark extension now; choose A, B or C at the freeze. My recommendation today is B: every cell is run on both models, and the frontier model carries half the files.
2. The fourth regime. Proposal: air-passenger rights under Regulation (EC) 261/2004 (delay, cancellation, denied boarding). Alternative: warranty for defective goods under the Code of Obligations, arts. 197 to 210.
3. Fix (d) changes what the artefact is: it stores readings and reuses them. I think the thesis asks for it ("code decides" includes deciding what need not be read again). Say if you see it differently.
4. Product frames for "what the system is" must come from the product session; I will write the request when you confirm the five items.
5. One north star I could not read: the OpenAI page refuses automated readers. If you want its structure followed closely, paste the text or a PDF of it.